Virtualization -- Guided Notes ANSWER KEY

Unit tp-2.10 | CompTIA Network+ N10-009 Obj. FC0-U71 2.10 | N10-008 FC0-U71 2.10

For instructor use / distribute after students complete the notes.

Question 1

Virtualization allows one physical machine (the _______) to run multiple simulated environments called _______ (VMs). A _______ is the software layer that creates and manages VMs. Examples include VMware vSphere, Microsoft Hyper-V, and _______.

Answer: Host; virtual machines; hypervisor; VirtualBox (or KVM).

Question 2

A Type 1 hypervisor runs _______ on bare metal hardware, without a host OS underneath. Examples include VMware _______ and Microsoft _______. A Type 2 hypervisor runs on top of an existing _______ and is used on workstations. Examples include _______ and VirtualBox.

Answer: Directly; ESXi; Hyper-V; OS; VMware Workstation.

Question 3

Server consolidation uses virtualization to run many VMs on fewer _______ machines. If a company runs 20 servers at 5% CPU utilization each, they waste _______ capacity. Consolidating them onto fewer powerful physical hosts improves _______ utilization and reduces hardware, power, and _______ costs.

Answer: Physical; 95%; resource; cooling.

Question 4

A VM _______ is a saved state of a VM at a point in time. If a software update or configuration change breaks the VM, the administrator can _______ to the snapshot and restore the prior state. This is a powerful feature for _______ environments where changes carry risk.

Answer: Snapshot; revert (roll back); test/staging.

Question 5

Containers (like _______) differ from VMs because they share the _______ kernel of the host OS rather than running a separate OS per instance. Containers are _______ (more/less) lightweight and start _______ (faster/slower) than VMs. They are commonly used in microservices and cloud deployments.

Answer: Docker; OS (host); more; faster.

Question 6

VDI (Virtual Desktop Infrastructure) hosts users _______ on a central server. End users connect with _______ clients or thin clients. The advantage is that data and processing stay in the _______, making it secure and easy to manage. The risk is that if the server goes down, _______ users lose access.

Answer: Desktop environments (operating system desktops); remote desktop; data center; all.

Question 7

VM isolation means that a _______, crash, or attack inside one VM does not _______ other VMs running on the same host (under normal conditions). Each VM has _______ virtual hardware resources. This containment is what makes virtualization valuable for running _______ or testing malware.

Answer: Virus; affect; isolated; sandboxing.

Question 8 -- Real World Application

REAL WORLD: A cybersecurity teacher wants students to practice using Kali Linux for penetration testing exercises. The school computers run Windows 10 and cannot be reimaged. The teacher does not want students reachable from the school network during labs. Explain how virtualization solves all three constraints and describe the specific configuration you would set up.

Answer: Solution: Install VirtualBox (Type 2 hypervisor) on each Windows 10 machine. Create a Kali Linux VM inside VirtualBox on each student computer. This solves all three constraints: (1) No reimaging needed -- Kali runs inside Windows as a guest VM; the host Windows system is unaffected and students can switch between Kali and Windows. (2) Network isolation -- configure each Kali VM with a Host-Only or NAT network adapter instead of a Bridged adapter. Host-Only creates a private network visible only to the host, preventing the VM from reaching the school network or being reached by other machines on the LAN. (3) Repeatable clean state -- create a base snapshot after initial setup so students can revert after each lab.